MSG No. 215.71 $P\bar{4}3m1'$ [Type II, cubic]

Table 1: Wyckoff site: `1a`, site symmetry: `-43m1'`

No.	position	mapping
1	[0, 0, 0]	`[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24,`
		`25,26,27,28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,44,45,46,47,48]`

Table 2: Wyckoff site: `1b`, site symmetry: `-43m1'`

No.	position	mapping
1	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24,`
		`25,26,27,28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,44,45,46,47,48]`

Table 3: Wyckoff site: `3c`, site symmetry: `-42.m1'`

No.	position	mapping
1	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[1,2,3,4,13,14,21,22,25,26,27,28,37,38,45,46]`
2	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[5,8,10,12,17,18,19,20,29,32,34,36,41,42,43,44]`
3	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[6,7,9,11,15,16,23,24,30,31,33,35,39,40,47,48]`

Table 4: Wyckoff site: `3d`, site symmetry: `-42.m1'`

No.	position	mapping
1	$\left[\frac{1}{2}, 0, 0\right]$	`[1,2,3,4,13,14,21,22,25,26,27,28,37,38,45,46]`
2	$\left[0, \frac{1}{2}, 0\right]$	`[5,8,10,12,17,18,19,20,29,32,34,36,41,42,43,44]`
3	$\left[0, 0, \frac{1}{2}\right]$	`[6,7,9,11,15,16,23,24,30,31,33,35,39,40,47,48]`

Table 5: Wyckoff site: `4e`, site symmetry: `.3m1'`

No.	position	mapping
1	$[x, x, x]$	`[1,5,6,20,22,24,25,29,30,44,46,48]`
2	$[x, -x, -x]$	`[2,10,11,16,17,21,26,34,35,40,41,45]`
3	$[-x, x, -x]$	`[3,7,12,13,18,23,27,31,36,37,42,47]`
4	$[-x, -x, x]$	`[4,8,9,14,15,19,28,32,33,38,39,43]`

Table 6: Wyckoff site: `6f`, site symmetry: `2.mm1'`

No.	position	mapping
1	$[x, 0, 0]$	`[1,2,21,22,25,26,45,46]`
2	$[-x, 0, 0]$	`[3,4,13,14,27,28,37,38]`
3	$[0, x, 0]$	`[5,12,18,20,29,36,42,44]`
4	$[0, 0, x]$	`[6,9,15,24,30,33,39,48]`
5	$[0, 0, -x]$	`[7,11,16,23,31,35,40,47]`

continued ...

Table 6

No.	position	mapping
6	$[0, -x, 0]$	[8,10,17,19,32,34,41,43]

Table 7: Wyckoff site: 6g, site symmetry: 2.mm1'

No.	position	mapping
1	$\left[x, \frac{1}{2}, \frac{1}{2}\right]$	[1,2,21,22,25,26,45,46]
2	$\left[-x, \frac{1}{2}, \frac{1}{2}\right]$	[3,4,13,14,27,28,37,38]
3	$\left[\frac{1}{2}, x, \frac{1}{2}\right]$	[5,12,18,20,29,36,42,44]
4	$\left[\frac{1}{2}, \frac{1}{2}, x\right]$	[6,9,15,24,30,33,39,48]
5	$\left[\frac{1}{2}, \frac{1}{2}, -x\right]$	[7,11,16,23,31,35,40,47]
6	$\left[\frac{1}{2}, -x, \frac{1}{2}\right]$	[8,10,17,19,32,34,41,43]

Table 8: Wyckoff site: 12h, site symmetry: 2..1'

No.	position	mapping
1	$\left[x, \frac{1}{2}, 0\right]$	[1,2,25,26]
2	$\left[-x, \frac{1}{2}, 0\right]$	[3,4,27,28]
3	$\left[0, x, \frac{1}{2}\right]$	[5,12,29,36]
4	$\left[\frac{1}{2}, 0, x\right]$	[6,9,30,33]
5	$\left[\frac{1}{2}, 0, -x\right]$	[7,11,31,35]
6	$\left[0, -x, \frac{1}{2}\right]$	[8,10,32,34]
7	$\left[-x, 0, \frac{1}{2}\right]$	[13,14,37,38]
8	$\left[0, \frac{1}{2}, x\right]$	[15,24,39,48]
9	$\left[0, \frac{1}{2}, -x\right]$	[16,23,40,47]
10	$\left[\frac{1}{2}, -x, 0\right]$	[17,19,41,43]
11	$\left[\frac{1}{2}, x, 0\right]$	[18,20,42,44]
12	$\left[x, 0, \frac{1}{2}\right]$	[21,22,45,46]

Table 9: Wyckoff site: 12i, site symmetry: ..m1'

No.	position	mapping
1	$[x, x, z]$	[1,20,25,44]
2	$[x, -x, -z]$	[2,17,26,41]
3	$[-x, x, -z]$	[3,18,27,42]
4	$[-x, -x, z]$	[4,19,28,43]
5	$[z, x, x]$	[5,24,29,48]
6	$[x, z, x]$	[6,22,30,46]
7	$[-x, z, -x]$	[7,13,31,37]
8	$[-z, -x, x]$	[8,15,32,39]
9	$[-x, -z, x]$	[9,14,33,38]
10	$[z, -x, -x]$	[10,16,34,40]
11	$[x, -z, -x]$	[11,21,35,45]
12	$[-z, x, -x]$	[12,23,36,47]

Table 10: Wyckoff site: `24j`, site symmetry: `11'`

No.	position	mapping
1	$[x, y, z]$	`[1,25]`
2	$[x, -y, -z]$	`[2,26]`
3	$[-x, y, -z]$	`[3,27]`
4	$[-x, -y, z]$	`[4,28]`
5	$[z, x, y]$	`[5,29]`
6	$[y, z, x]$	`[6,30]`
7	$[-y, z, -x]$	`[7,31]`
8	$[-z, -x, y]$	`[8,32]`
9	$[-y, -z, x]$	`[9,33]`
10	$[z, -x, -y]$	`[10,34]`
11	$[y, -z, -x]$	`[11,35]`
12	$[-z, x, -y]$	`[12,36]`
13	$[-x, z, -y]$	`[13,37]`
14	$[-x, -z, y]$	`[14,38]`
15	$[-z, -y, x]$	`[15,39]`
16	$[z, -y, -x]$	`[16,40]`
17	$[y, -x, -z]$	`[17,41]`
18	$[-y, x, -z]$	`[18,42]`
19	$[-y, -x, z]$	`[19,43]`
20	$[y, x, z]$	`[20,44]`
21	$[x, -z, -y]$	`[21,45]`
22	$[x, z, y]$	`[22,46]`
23	$[-z, y, -x]$	`[23,47]`
24	$[z, y, x]$	`[24,48]`